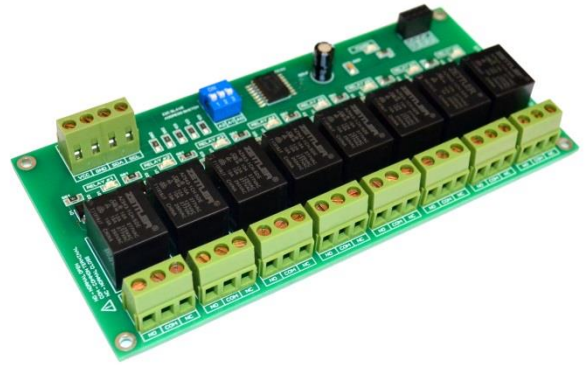


8 Channel Electromagnetic relay module with I²C controlling interface



8 Channel electromagnetic board has been designed for easy inductive and resistive load switching via I²C communication protocol. For wiring this board is needed only 4 wires - 2 wires with data lines SCL (serial clock) and SDA (serial data) and 2 wires with power supply VCC and GND. On board is special I²C slave address switch, which help to select board slave address. Total available eight slave addresses, therefore on one I²C line can be connected 8 relays modules. That mean user can control each of 64 relay separately. Please note, can't be connected two board with the same addresses on the I²C line. All I²C slave addresses should be different. For the all available I²C slave addresses watch table below.

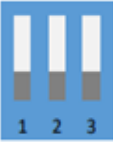
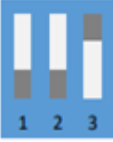
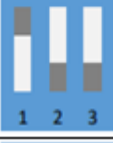
This 8 channel I²C electromagnetic module is fully compatible with any microcontroller (AVR, PIC, ARM, STM32), popular platform of Arduino and Raspberry, Wi-Fi ESP8266 and ESP32, and with other microcontrollers with I²C interface. I²C communication speed is up to 100kHz. Module recommended power supply is 5.0V DC, but also it can work at the lower power supply 3.3V DC. Power supply current should be at least 700mA @ 5V. In this module are used high quality electromagnetic relays ZETTLER. These relays have SPDT switching terminals and can switch load current up to 10A at DC and AC voltages. Switching load voltage range for AC is 0...240V and for DC range is 0...100V. For safe switching load wires connection are used quality screw terminals KF-128-3P. Each relay is indicated with led (color red).

Device description:

- PCB board dimension 72mm x 150mm
- PCB board material is FR-4 1.6mm, with solder mask and silk screen
- Compatible with DIN rail plastic holder
- Operation DC voltage – 3.3V and 5.0V(recommended)
- Maximum current @5.0V – 700mA
- Maximum current per relay – 10A peak
- LEDs indicators for each electromagnetic relay channel
- Relays with SPDT switch terminals: Normally Open, Common, Normally Closed
- Relay contact resistance 100 milliohms max. (initial value)
- Relay insulation resistance 100 MOhm min. (DC 500V)
- Relay operation time 8 ms max
- Relay release time 5ms max
- Relay dielectric strength 750 VAC, 50/60Hz between contact
- Relay dielectric strength 1,500 VAC, 50/60Hz between all elements
- Relay expected life Mechanical - 10,000,000 operations min.
- Relay expected life Electrical - 100,000 operations min. at rated load
- Working temperature range - 25 C ~ + 80 C
- Comes with different I²C slave address 0x27 or 0x3F, depends on the batch
- The module contains an 8bit I²C expander PCF8574 or PCF8574A chip
- Each board can be assigned an I²C address between 0x20...0x27 (0x38...0x3F) by the DIP switch
- The logic power supply voltage should match the voltage levels on the I²C bus
- The SDA and SCL lines are pulled up to VCC with 5.6k resistors on the relay module
- Up to 8 boards can be connected to one I²C line

I²C slave addresses switch description:



	PCF8574 I2C-bus slave address	PCF8574A I2C-bus slave address
	0x27	0x3F
	0x26	0x3E
	0x25	0x3D
	0x24	0x3C
	0x23	0x3B
	0x22	0x3A
	0x21	0x39
	0x20	0x38

I²C controlling:

This relay module is based on the I²C 8bit PCF8574(A) I/O I2C expander. Chip should be setup to the output mode. Each relay is controlled via logic levels 0 and 1 in the corresponding bit.

MSB							LSB	
1	1	1	1	1	1	1	0	8th relay is ON
1	0	1	1	1	1	1	0	2nd and 8th relays are ON
1	0	1	0	1	0	1	1	2nd, 4th and 6th relays are ON
0	1	1	1	0	0	1	0	1st, 5th, 6th and 8th relays are ON
0	0	1	0	1	1	1	1	1st, 2nd and 4th relays are ON

C/C++ Arduino IDE code examples:

```
Wire.beginTransmission(0x27); // slave address 0x27
Wire.write(0b11111110);      // 8th relay is ON
Wire.endTransmission();

Wire.beginTransmission(0x27); // slave address 0x27
Wire.write(0b10111110);      // 2nd and 8th relays are ON
Wire.endTransmission();

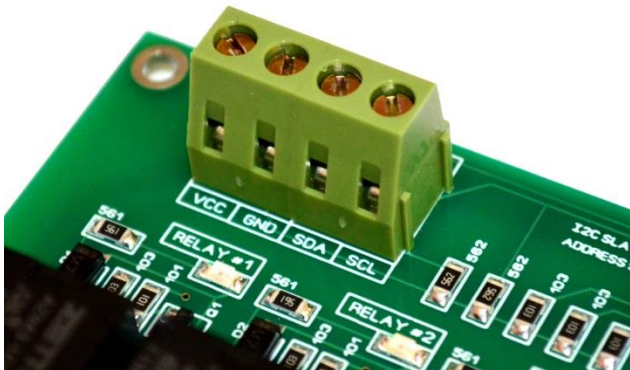
Wire.beginTransmission(0x27); // slave address 0x27
Wire.write(0b10101011);      // 2nd, 4th and 6th relays are ON
Wire.endTransmission();

Wire.beginTransmission(0x27); // slave address 0x27
Wire.write(0b01110010);      // 1st, 5th, 6th and 8th relays are ON
Wire.endTransmission();

Wire.beginTransmission(0x27); // slave address 0x27
Wire.write(0b00101111);      // 1st, 2nd and 4th relays are ON
Wire.endTransmission();
```

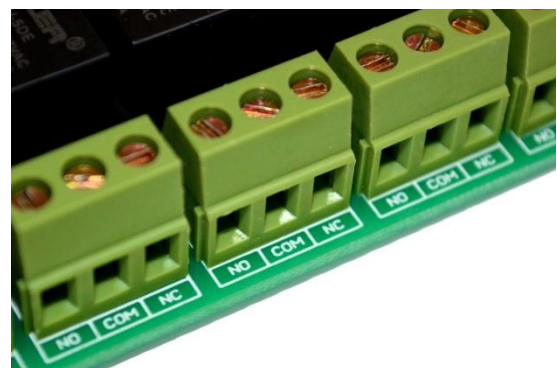
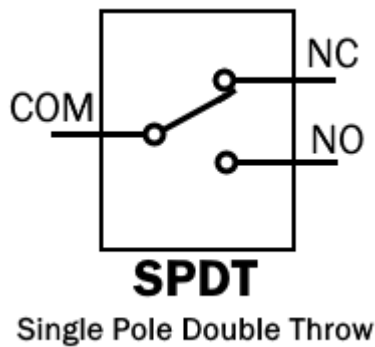
Screw terminal description:

- **VCC** – device power supply (+)
- **GND** - device ground (-)
- **SDA** – I²C serial data
- **SCL** – I²C serial clock

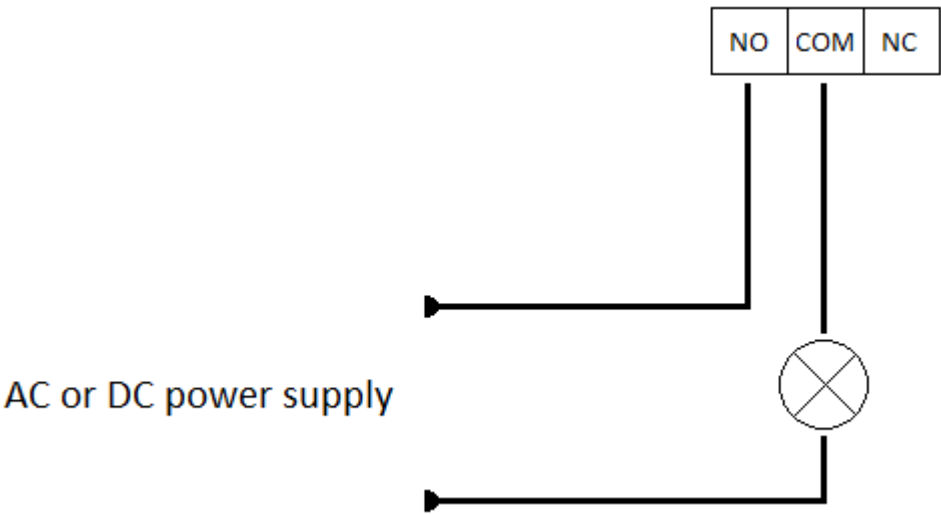


Switching relay description:

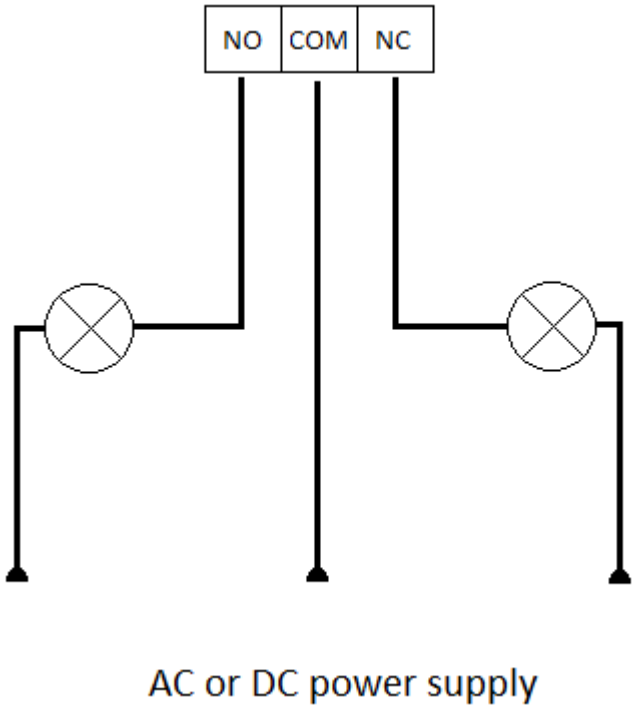
COM – common terminal
NC – normally closed
NO – normally opened



Single load wiring ON/OFF:

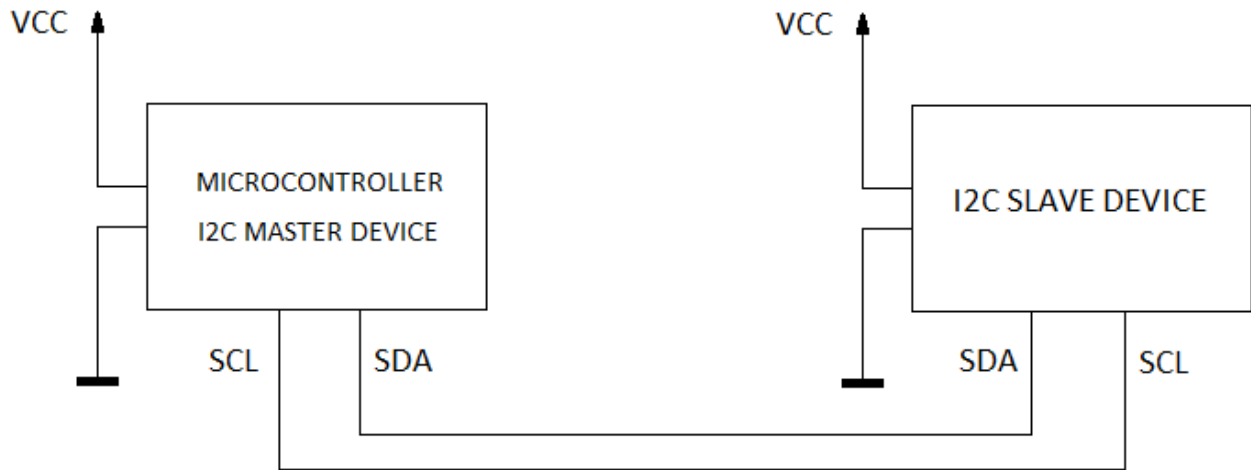


Double load wiring for switching:

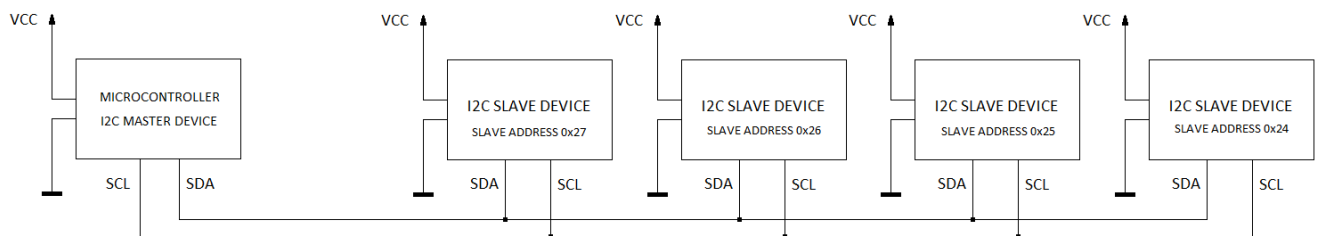


I²C wiring master – slave:

Single board wiring:



Multi boards wiring:



PCB dimension:

